

Equation (17) also shows exactly how to weight together averages of x_{it} and perform a single t -test that directly captures the identifying assumption. Generate a dummy for the effective treatment group, $B_k = w_k^T - w_k^C > 0$, then regress timing-group means, $\bar{x}_k$, on B_k weighting by $|w_k^T - w_k^C|$. The coefficient on B_k equals covariate differences weighted by the actual identifying variation, and its t -statistic tests the null of reweighted balance in (17). One can also use this strategy to test for pre-treatment trends in confounders (or the outcome) by regressing $\bar{x}_{kt}$ on B_k , year dummies, and their interaction, or the interaction of B_k with a linear trend using dates before any treatment starts.²⁶

The reweighted balance test has advantages over other strategies for testing balance in this setting. Regressing x_{it} on a constant and dummies for timing groups allows a test of the null of joint balance across groups ($H_0: \bar{x}_k - \bar{x}_U = 0, \forall k \in K$). With many timing groups, however, this F -test will have low power, it does not reflect the importance of each group ($w_k^T - w_k^C$), and does not show the sign or magnitude of any imbalance.²⁷ The reweighted test, on the other hand, has higher power and describes the sign and magnitude of imbalance. It is also better than a test for linear relationships between x_{it} and t_k^* or tests for balance between “early” and “late” treated units (Almond, Hoynes, and Schanzenbach 2011, Bailey and Goodman-Bacon 2015). Because the effective control group can include *both* the earliest and latest treated units, failing to find a linear relationship between $\bar{x}_k$ and t_k^* can miss the relevant imbalance between the most important “middle” units and the end points.

²⁶ Plotting confounders or pre-trends across groups, however, is important to ensure that a failure to reject does not reflect offsetting trends or covariate means across timing groups.

²⁷ Note that using an event-study specification to evaluate pre-trends does not test for joint equality of pre-trends either. It tests for common trends using the (heretofore unarticulated) estimator itself.

III. DD DECOMPOSITION IN PRACTICE: UNILATERAL DIVORCE AND FEMALE SUICIDE

To illustrate how to use DD decomposition theorem in practice, I replicate Stevenson and Wolfers' (2006) analysis of no-fault divorce reforms and female suicide. Unilateral (or no-fault) divorce allowed either spouse to end a marriage, redistributing property rights and bargaining power relative to fault-based divorce regimes. Stevenson and Wolfers exploit "the natural variation resulting from the different timing of the adoption of unilateral divorce laws" in 37 states from 1969-1985 (see table 1) using the "remaining fourteen states as controls" to evaluate the effect of these reforms on female suicide rates. Figure 5 replicates their event-study result for female suicide using an unweighted model with no covariates.²⁸ Our results match closely: suicide rates display no clear trend before the implementation of unilateral divorce laws, but begin falling soon after. They report a DD coefficient in logs of -9.7 (s.e. = 2.3). I find a DD coefficient in levels of -3.08 (s.e. = 1.13), or a proportional reduction of 6 percent.²⁹

A. Describing the design

Figure 6 uses the DD decomposition theorem to illustrate the sources of variation. I plot each 2x2 DD against its weight and calculate the average effect and total weight for the three types of 2x2 comparisons: treated/untreated, early/late, late/early.³⁰ As theorem 1 states, the two-way fixed effects estimate, -3.08, is an average of the y-axis values weighted by their x-axis values. Summing the weights on timing terms ($s_{k\ell}$) shows exactly how much of $\hat{\beta}^{DD}$ comes from timing variation (37 percent). The large untreated group puts a lot of weight on $\hat{\beta}_{kU}^{2x2}$ terms, but more on those

²⁸ Data on suicides by age, sex, state, and year come from the National Center for Health Statistics' Multiple Cause of Death files from 1964-1996, and population denominators come from the 1960 Census (Haines and ICPSR 2010) and the Surveillance, Epidemiology, and End Results data (SEER 2013). The outcome is the age-adjusted (using the national female age distribution in 1964) suicide mortality rate per million women. The average suicide rate in my data is 52 deaths per million women versus 54 in Stevenson and Wolfers (2006). My replication analysis uses levels to match their figure, but the conclusions all follow from a log specification as well.

²⁹ The differences in the magnitudes likely come from three sources: age-adjustment (the original paper does not describe an age-adjusting procedure); data on population denominators; and my omission of Alaska and Hawaii.

³⁰ There are 156 distinct DD components: 12 comparisons between timing groups and pre-reform states, 12 comparisons between timing groups and non-reform states, and $(12^2 - 12)/2 = 66$ comparisons between an earlier switcher and a later non-switcher, and 66 comparisons between a later switcher and an earlier non-switcher

involving pre-1964 reform states (38.4 percent) than non-reform states (24 percent). Figure 6 also highlights the role of a few influential 2x2 terms—comparisons between the 1973 states and non-reform or pre-1964 reform states account for 18 percent of the estimate, and the ten highest-weight 2x2 DD’s account for over half.

The bias resulting from time-varying effects is also apparent in figure 6. The average of these post-treatment event-study estimates in figure 5 is -4.92, while the DD estimate is just 60 percent as large. The difference stems from the comparisons of later- to earlier-treated groups. The average treated/untreated estimates are negative (-5.33 and -7.04) as are the comparisons of earlier- to later-treated states (although less so: -0.19).³¹ The comparisons of later- to earlier-treated states, however, are *positive* on average (3.51) and account for the bias in the overall DD estimate. Using the decomposition theorem to take these terms out of the weighted average yields an effect of -5.44—close to the average of the event-study coefficients. The DD decomposition theorem shows that one way to summarize effects in the presence of time-varying heterogeneity is simply to subtract the components of the DD estimate that are biased using the weights in equation (7).

B. Testing the design

Figures 7 and 8 test for covariate balance in the unilateral divorce analysis. Figure 6 plots the reweighted balance test weights, $w_k^T - w_k^C$, from equation (17), the corresponding weights from a timing-only design, and each group’s sample share. Larger groups have larger weight, but because they have relatively low treatment variance, the earliest timing groups are downweighted relative to their sample shares.³² In fact, the 1969 states effectively act as controls because $w_k^T - w_k^C < 0$.

³¹ This point also applies to units that are already treated at the beginning of the panel, like the pre-1964 reform states in the unilateral divorce analysis. Since their $\bar{D}_k = 1$ they can only act as an already-treated control group. If the effects for pre-1964 reform states were constant they would not cause bias.

³² Adding $5 \times \text{year}$ to the suicide rate for the 1970 states ($w_k^T - w_k^C = 0.0039$) changes the DD estimate from -3.08 to -2.75, but adding it to the 1973 group ($w_k^T - w_k^C = 0.18$) yields a very biased DD estimate of 12.28.

Figure 8 implements both a joint balance test and the reweighted test using two potential determinants of marriage market equilibria in 1960: per-capita income and the male/female sex ratio. Panel A shows that average per-capita income in untreated states (\$13,431) is lower than the average in every timing group except for those that implemented unilateral divorce in 1969 (which actually get more weight as controls) or 1985. The joint F -test, however, fails to reject the null hypothesis that these means are the same. It is not surprising that such a low power test (12 restrictions on 48 states observations) fails to generate strong evidence against the null. The reweighted test, on the other hand, does detect a difference in per-capita income of \$2,285 between effective treatment states—those that implemented unilateral divorce in 1970 or later—and effective control states—pre-1964 reform states, non-reform states, and the 1969 states. Panel B shows that the 1960 sex ratio is higher in almost all treatment states than in the control states. While the joint test cannot reject the null of equal means, the reweighted test does reject reweighted balance ($p = 0.06$).³³

C. Evaluating alternative specifications

Researchers almost always estimate models other than (2) and use differences across specifications to evaluate internal validity (Oster 2016) or choose projects in the first place. The DD decomposition theorem suggests a simple way to understand why estimates change. Stacking the 2x2 DD's and weights from (7) into vectors, we can write $\hat{\beta}^{DD} = \mathbf{s}'\hat{\beta}^{2x2}$. Any alternative specification that equals a weighted average can be written similarly, and the difference between

³³ One can run a joint test of balance across covariates using a seemingly unrelated regressions (SUR), as suggested by Lee and Lemieux (2010) in the regression discontinuity context. The results of these χ^2 tests are displayed at the top of figure 6. As with the separate balance tests, I fail to reject the null of equal means across groups and covariates. The joint reweighted balance test, however, does reject the null of equal weighted means between effective treatment and control groups. With 48 states and 12 timing groups, there are not sufficient degrees of freedom to implement a full joint test across many covariates. This is an additional rationale for the reweighted test.